

CS & IT ENGINEERING



Computer Network

Flow Control

Lecture No. - 03

By - Abhishek Sir





Recap of Previous Lecture



Topic

Stop-and-Wait Protocol





Topics to be Covered



Topic

Stop-and-Wait Protocol

ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

Telegram Link : https://t.me/abhisheksirCS_PW





Topic : Stop-and-Wait ARQ



Transmitter Protocol :

- Transmitter transmit one frame (with sequence number)
and wait for ACK of it until time-out
- After time-out, transmitter retransmit the frame (same sequence number)
and wait for ACK of it until time-out
- Transmitter transmit next frame (with alternate sequence number)
only after receiving ACK of transmitted frame



Topic : Stop-and-Wait ARQ



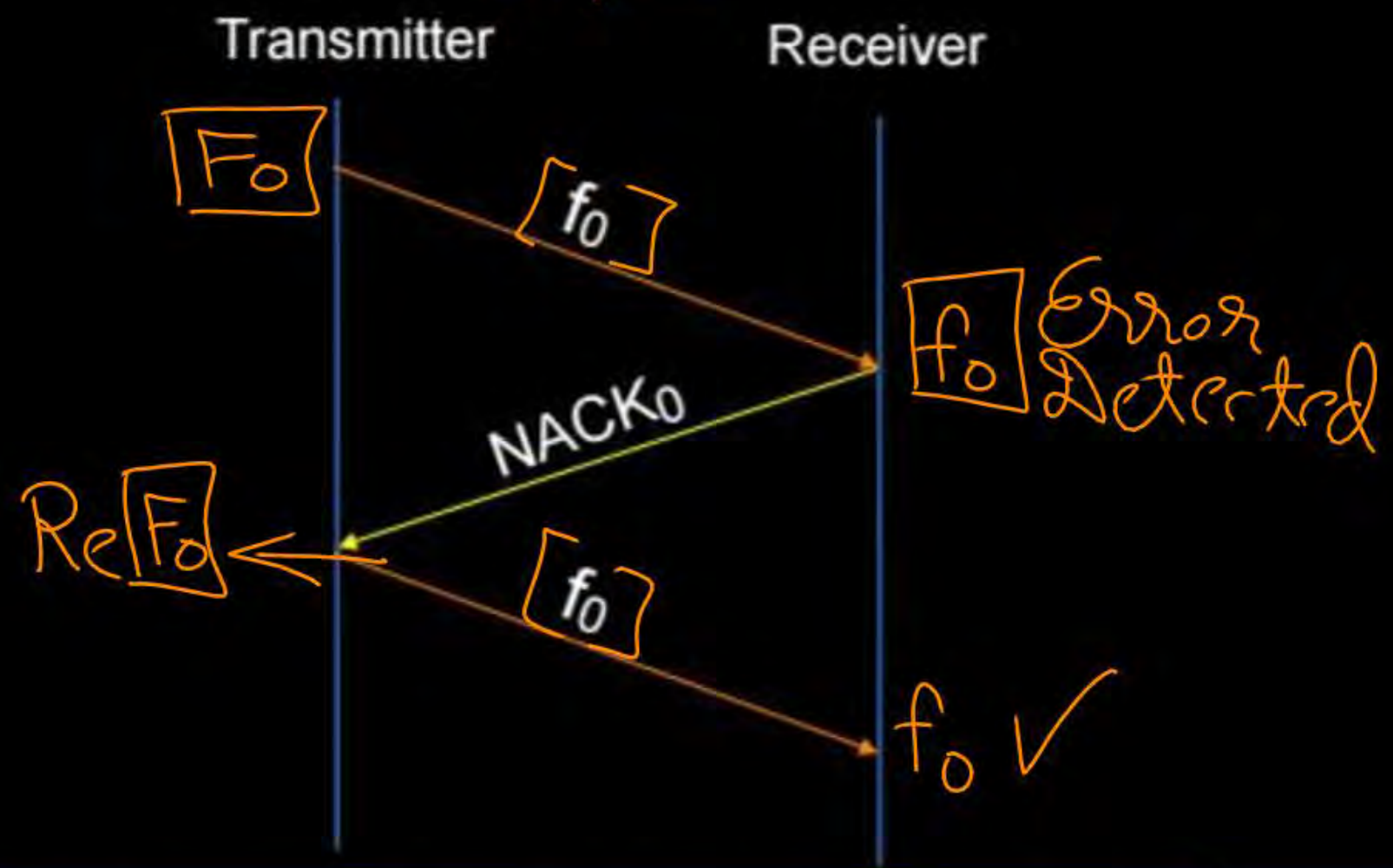
Receiver Protocol :

- Receiver transmit acknowledgment for every received frame after processing
- Acknowledgments carry corresponding frame sequence number

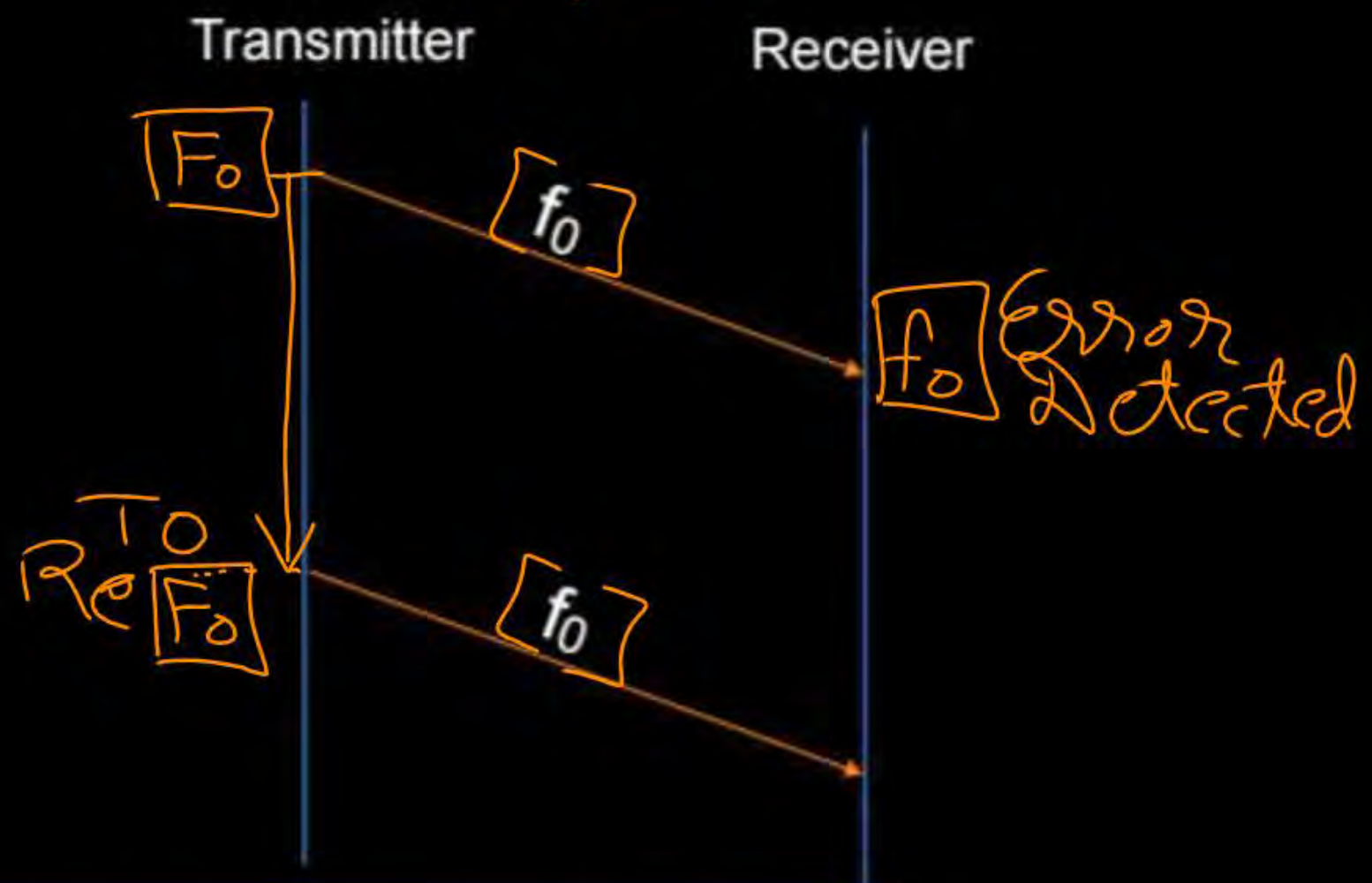


Topic : Stop-and-Wait ARQ

CASE I



CASE II



- | | |
|---------------------------------------|--|
| <u>ACK (positive acknowledgment)</u> | : <u>Leads next frame transmission</u> |
| <u>NACK (Negative acknowledgment)</u> | : <u>Leads retransmission of the frame</u> |
| <u>Time-out [ARQ]</u> | : <u>Leads retransmission of the frame</u> |



Topic : Stop-and-Wait ARQ

→ Transmitter's transmitting window size = 1

→ Receiver's receiving window size = 1

→ Total number of sequences = 2

[zero or ONE]



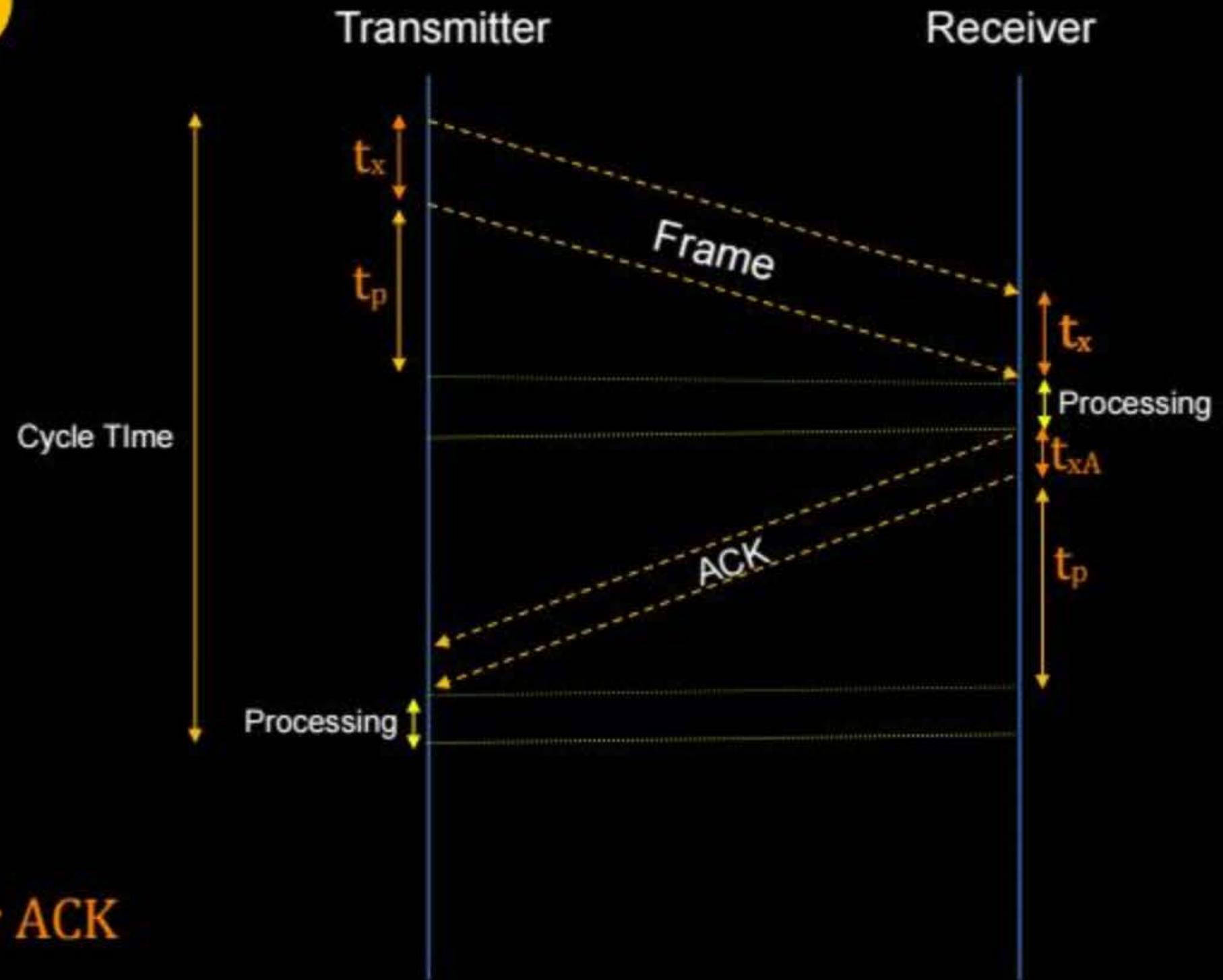
Topic : Cycle Time



$$\begin{aligned} \text{Cycle time} = & \text{Transmission delay} + \text{Propagation delay} \\ & + \text{Queuing delay at receiver} + \text{Processing delay by receiver for frame} \\ & + \text{Transmission delay for ACK} + \text{Propagation delay} \\ & + \text{Queuing delay at transmitter} + \text{Processing delay by transmitter for ACK} \end{aligned}$$



Topic : Cycle Time



t_{xA} : Transmission delay for ACK

$$t_{xA} \ll t_x$$



2 mins Summary



Topic

Stop-and-Wait ARQ



THANK - YOU

